

Assisted and Auto Bus Planning in Full-Chip Layout

Outline

- Introduction
- Assisted bus planning
- Auto bus planning
- Results
 - Tanglewood
 - Manzano
 - Nehalem

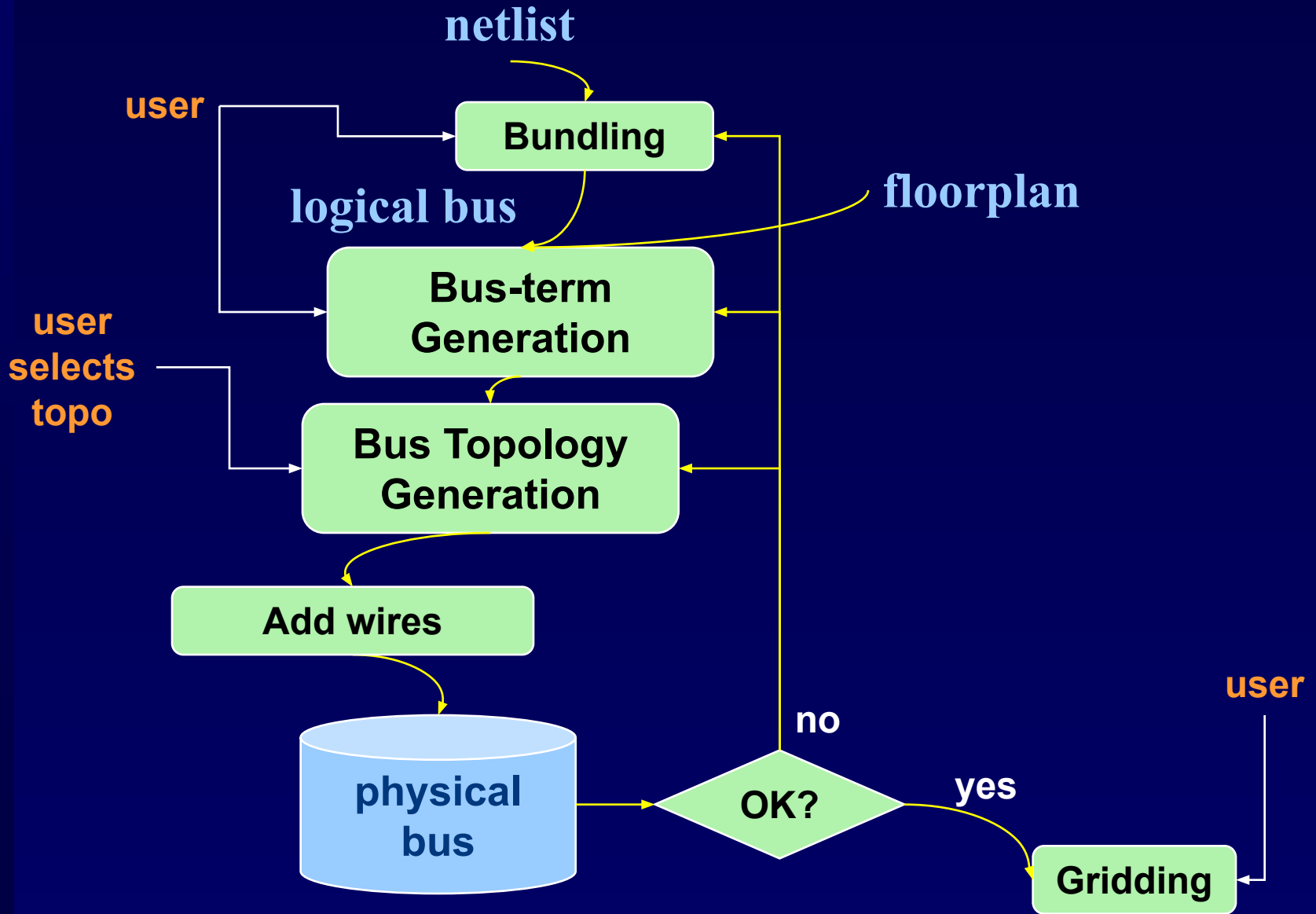
Introduction

- A set of new interconnect (IC) planning tools within Galaxy - DT's FCL tool
- Two objectives
 - Productivity in designing layout of critical IC
 - Plan full-chip: 2 weeks -> 1 day
 - Plan a unit: 2 days -> 10 minutes
 - Accurate top-down planning
 - Plan without layout/terminals
 - Plan down to lower levels of hierarchy
 - Manage congestion well

Bus Planning

- Input
 - Netlist and floor-plan
- Output
 - Interconnect as bus topologies
- Topology is not exact layout. But, it is close..
- Two major design flows:
 - Assisted: Help designers craft topologies
 - Auto: Craft and legalize topologies
- In design, use a combination of both

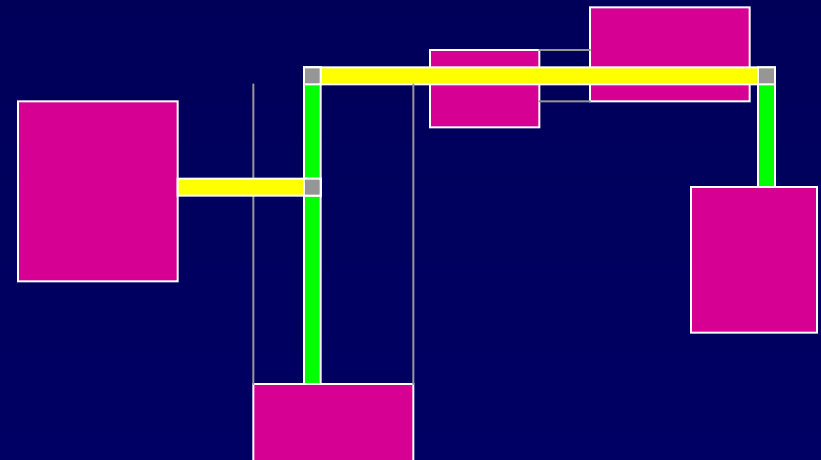
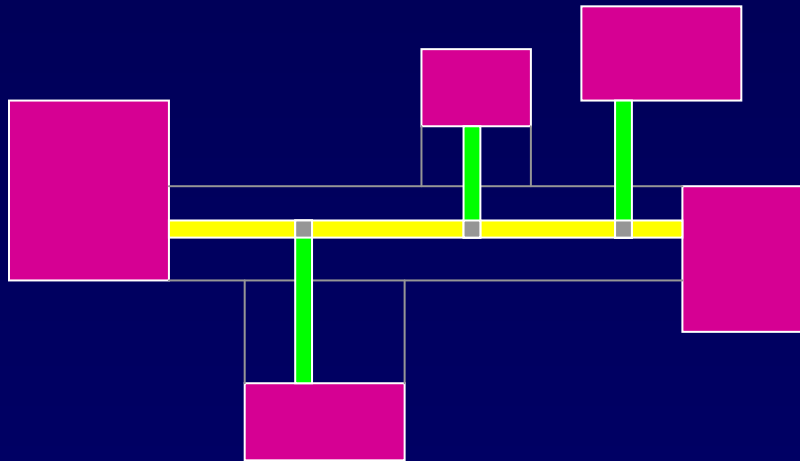
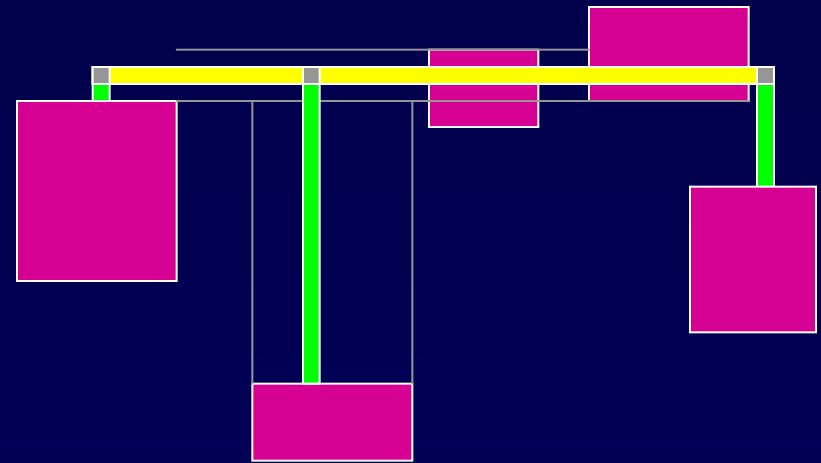
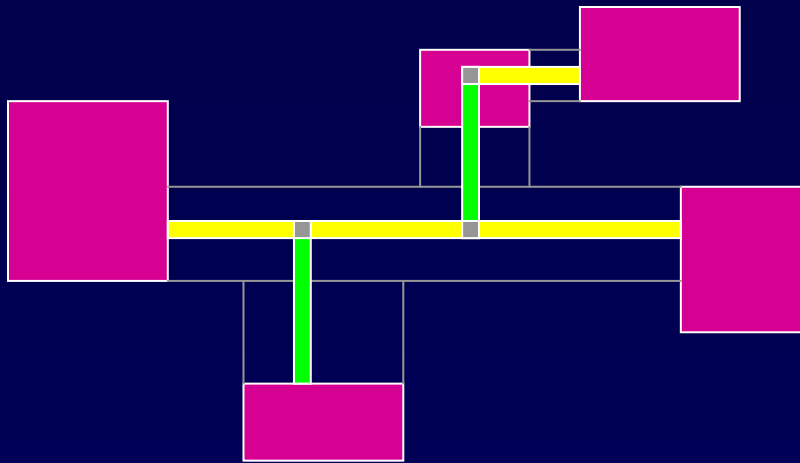
Assisted Bus Planning



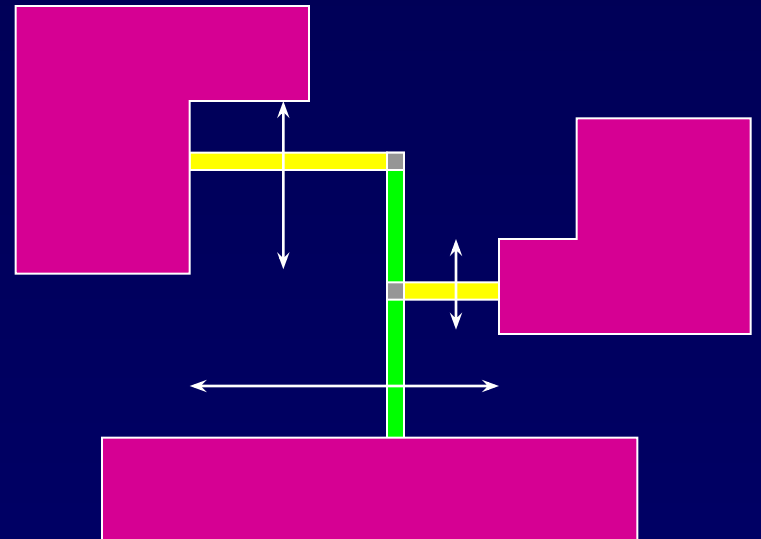
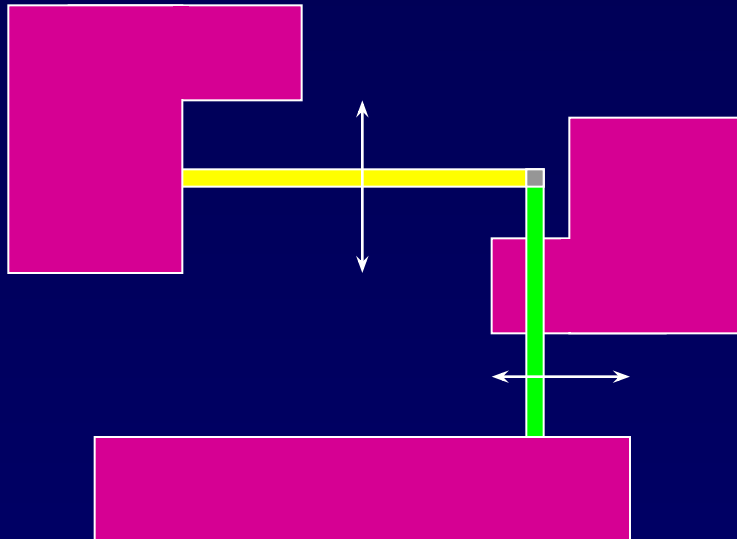
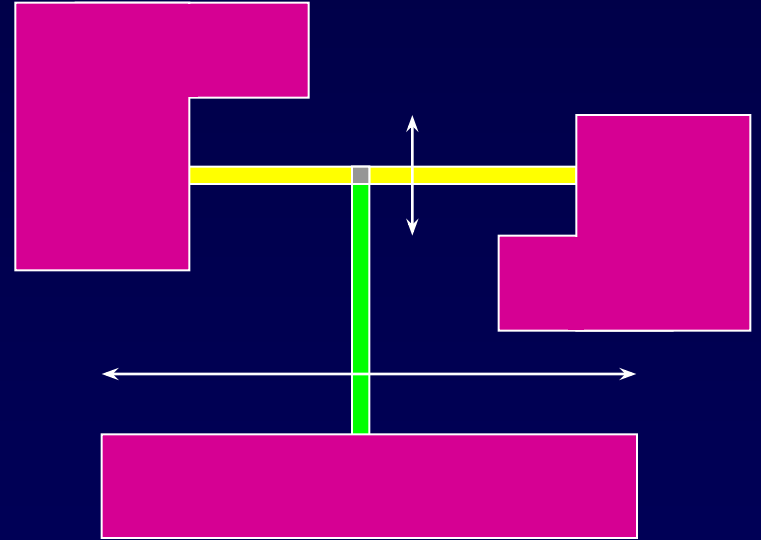
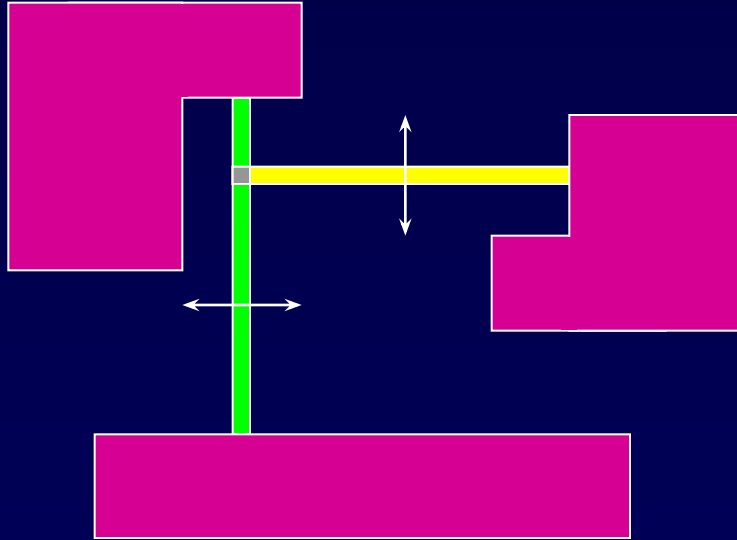
Topology Generation

- Goal
 - Design a set of good topologies
 - Do it fast: 5000 topologies per minute
- Input: bus-terms
- Output: set of topologies
- Each topology describes a unique path the bus can take from driver to receivers
- No exact layout or coords. No wires! Very fast!

Topology Alternatives



Topology Alternatives



Auto Bus Planning

- Builds on assisted bus planning
 - Legalize topologies (eliminate DRVs)
 - Choose topologies to reduce congestion
- Two major flows
 - Dilution based fast flow: 25k nets per minute
 - Obstacle aware flow: 1 bus per second

Dilution-based Flow

- Dilution: non-minimum spacing between wires
- Takes into account power/clock grid density
- Technique borrowed from Intel designers
- For early top-down design planning
- Provides accurate analysis of congestion
- Provides very fast feedback in order to:
 - Move blocks
 - Resize blocks/edges
 - Change bus topologies

Obstacle-aware Flow

- During placement of wires, avoid all obstacles
- Supports two design tasks
 - Plan buses in multiple hierarchy levels
 - Bus planning when power/pre-routes exist
 - Useful for late ECOs

Results

- Usage in projects
 - Tanglewood
 - Manzano
 - Nehalem
- Recent usage in Tejas and Whitefield

Tanglewood

- Data: A cpu core with three levels of hierarchy
 - box/section/fub
 - ~200 buses
- Used assisted bus planning
 - Plan buses at core to connect down to fubs
 - Select one topology per bus
 - Choose a location for each bus segment
 - Snap to grid to legalize

Tanglewood cont.

- Result
 - 95% of buses planned using topologies
- Testimony: “[Bus planning] was extremely helpful. Can’t imagine how long it would have taken us without it.”

Manzano

- Data: A flat cpu core of 25 blocks, 6k nets, ~300 buses
- Used dilution flow of auto bus planning to manage large RTL changes early in the design
- At each RTL drop, do:
 - Update SCH, logical buses
 - Assisted bus planning for 90% of buses
 - Run dilution flow
 - Move blocks/edges to reduce congestion

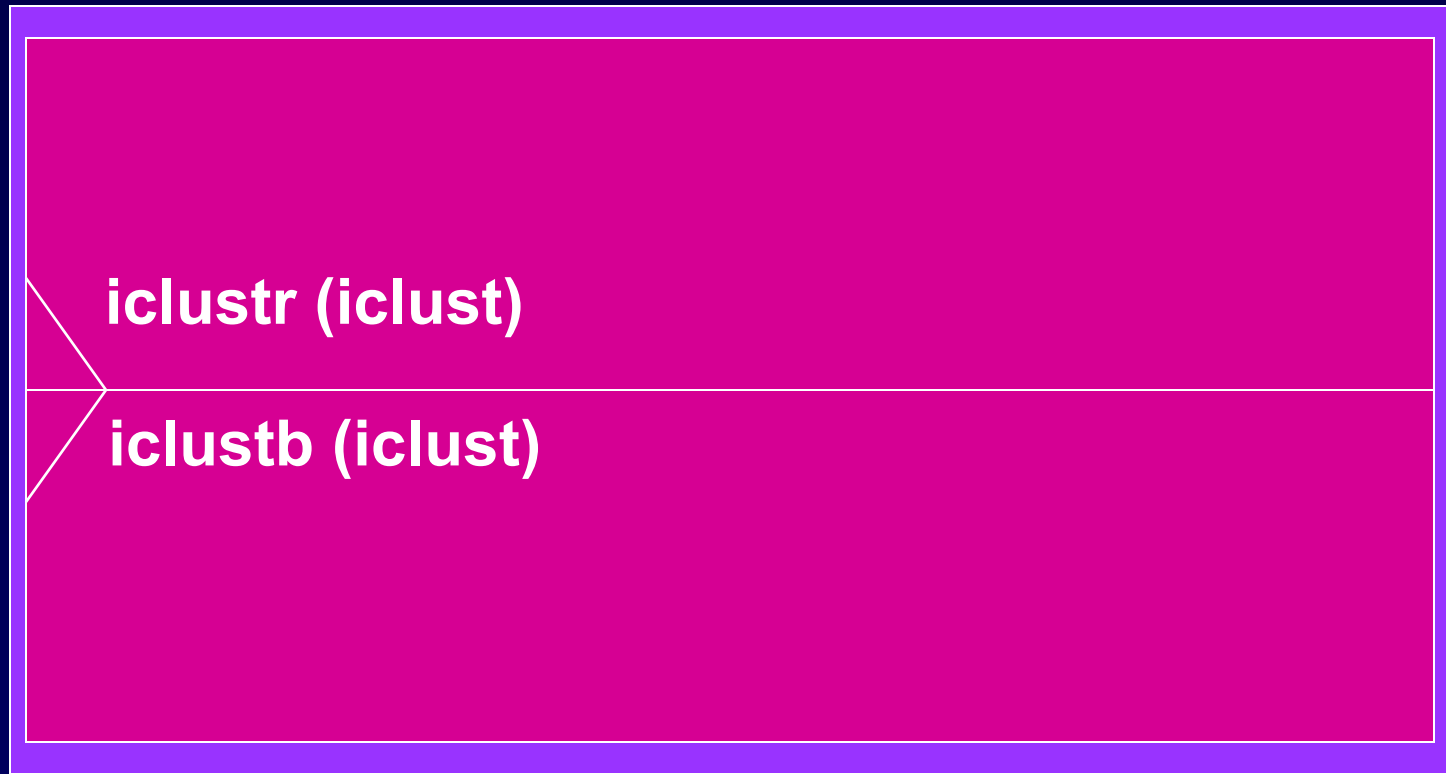
Manzano cont.

- Flow cont:
 - Re-generate all topologies and run dilution flow in minutes
 - While congested, visually analyze congestion, update floor-plan, re-run dilution flow
 - Snap segments to routing tracks using GRBUS in Galaxy
- Testimony: 2 weeks of bussing effort reduced to 1 day.

Nehalem

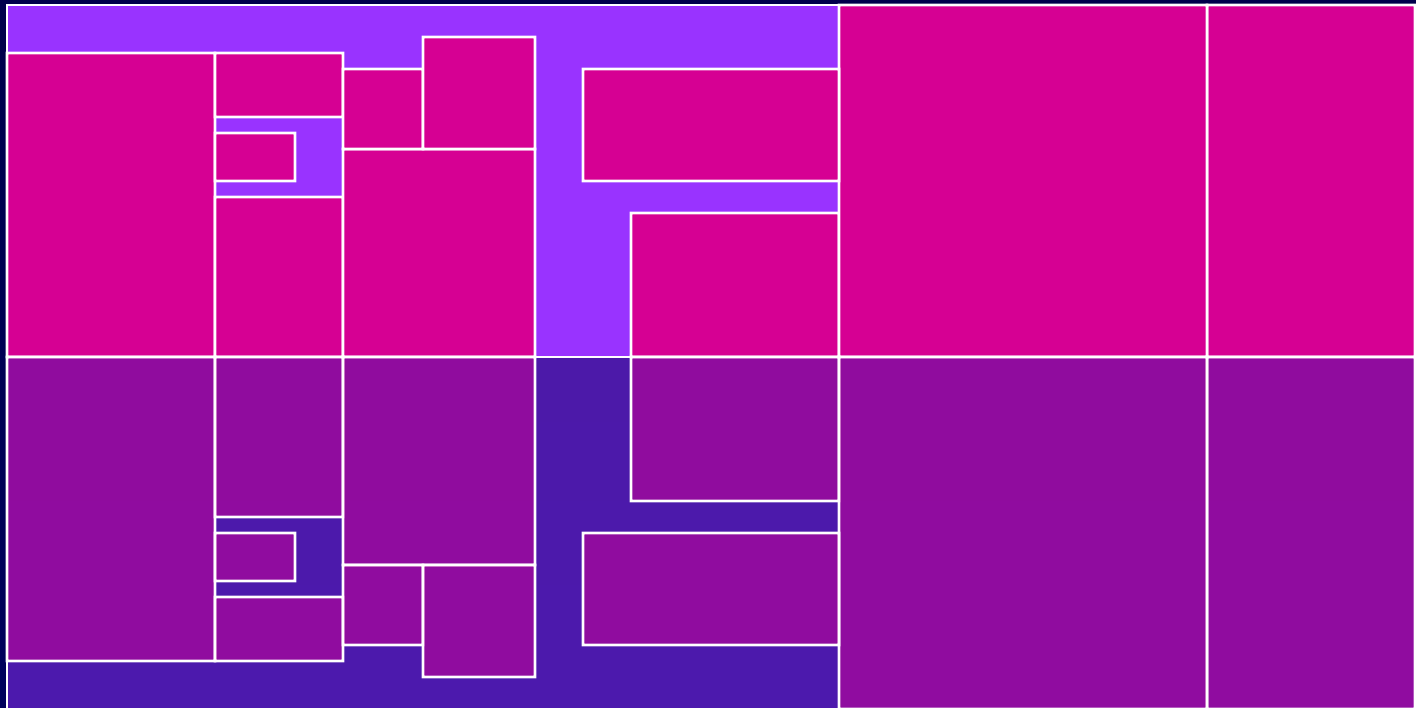
- Data: A unit (IEXEC) without terminals, with deep hierarchy (5 levels of blocks)
 - IEXEC contains two instances of ICLUST
 - Has ~2500 nets
 - ICLUST contains
 - 11 blocks, ~5000 nets
 - 4 levels of hierarchy

Nehalem IEXEC Depth=0

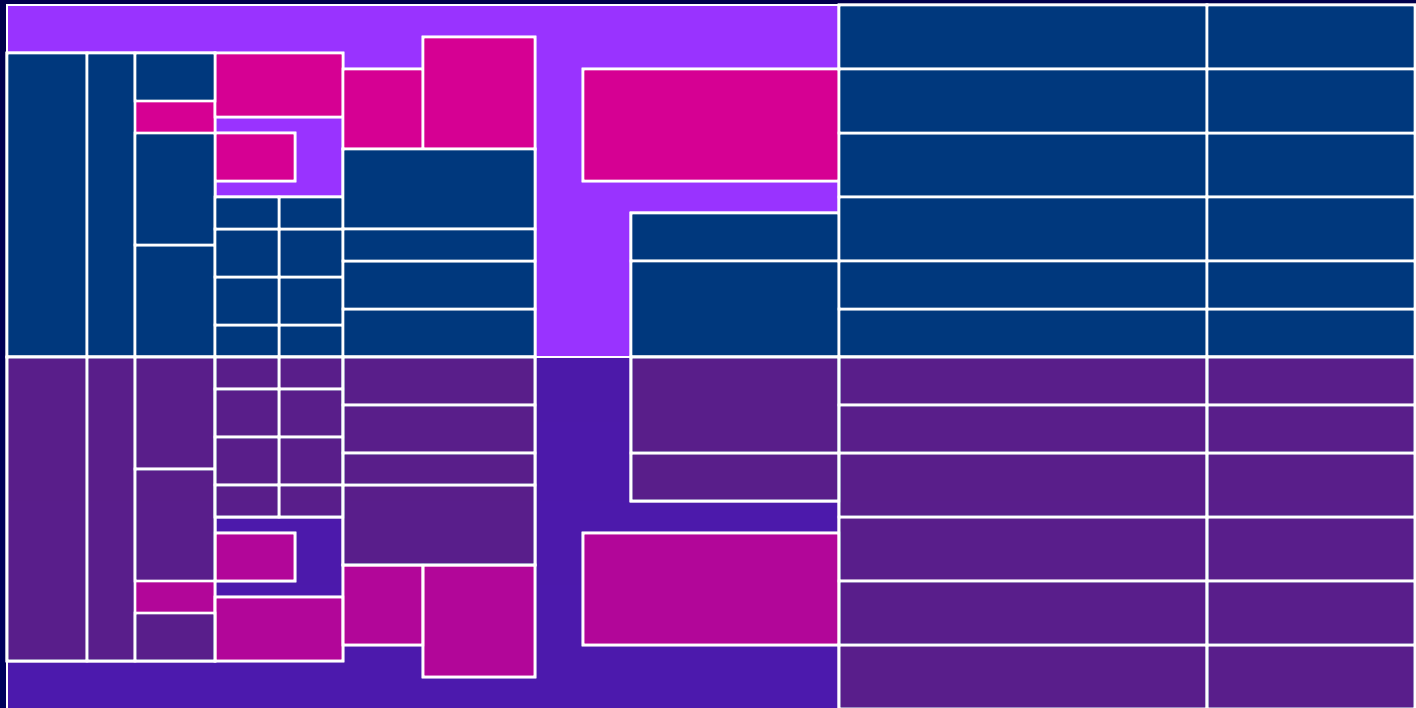


iexec

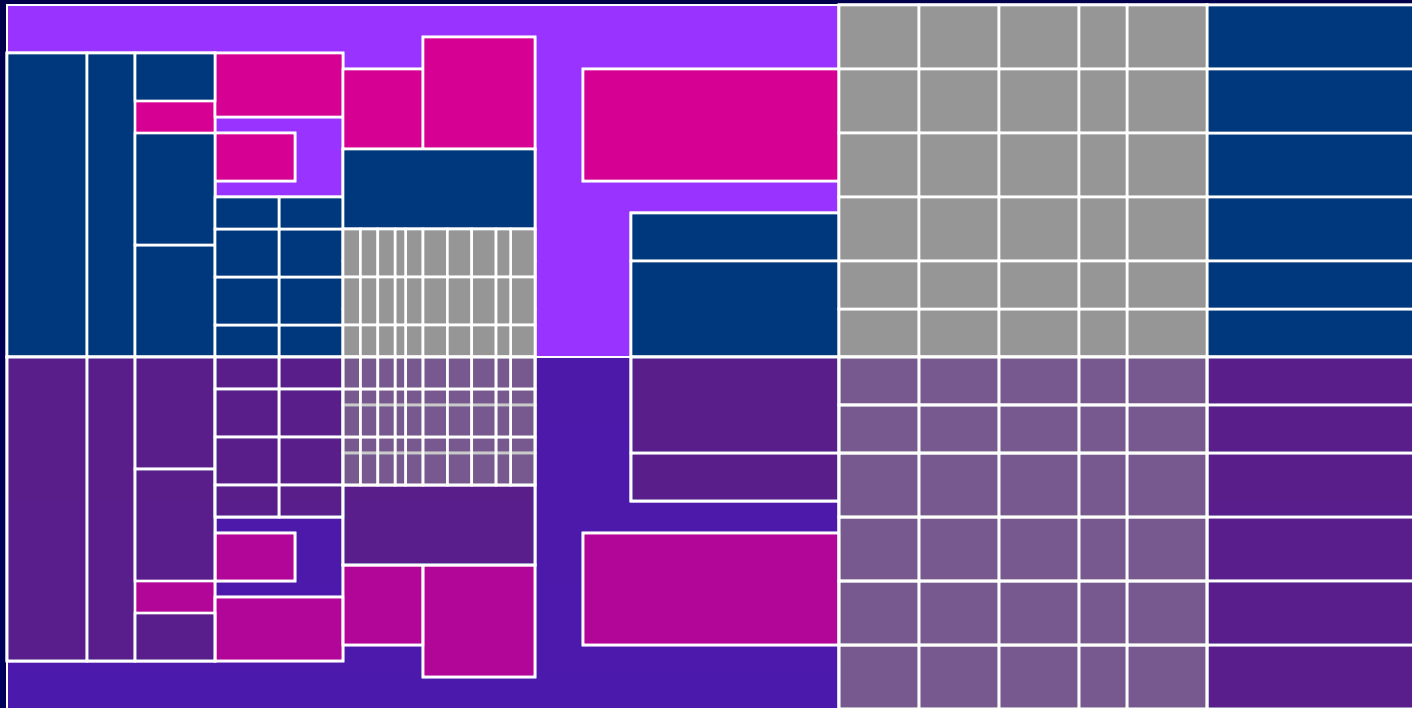
Nehalem ICLUST Depth=0



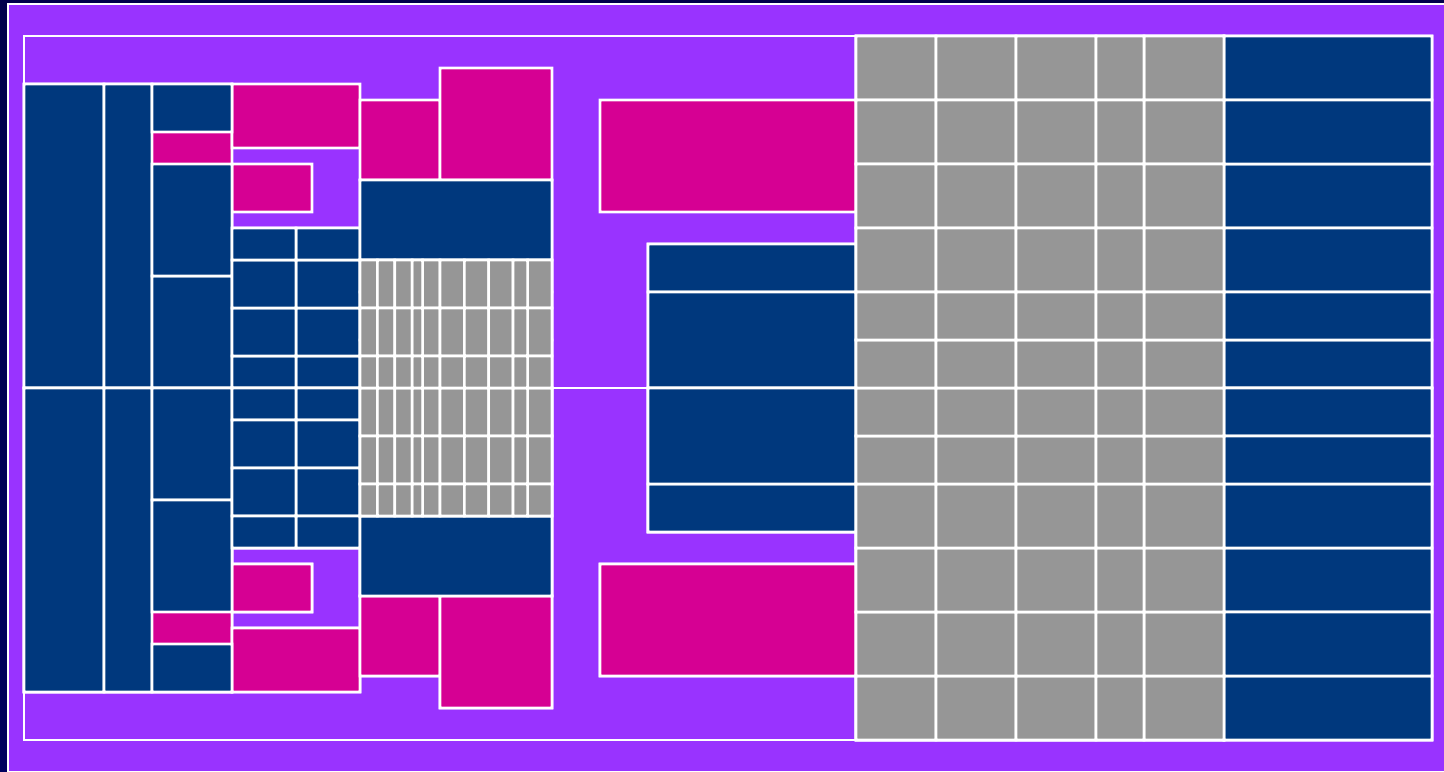
Nehalem ICLUST Depth=1



Nehalem ICLUST Depth=2



Nehalem IEXEC Depth=3



iexec

- Note: FUBs at depth=4 not shown

Nehalem cont.

- Used combination of dilution and obstacle-aware flows
 - First plan all buses local to ICLUST
 - Then, plan all IEXEC buses with buses from ICLUST as obstacles
- Designer testimony: **Planned all buses in 2 hours vs. an estimated 2-day effort of ChPPR-like manual bussing.**
- Used designers feedback to improve our tool
 - Today, total effort is only 10 minutes
- **See details in Galaxy Demo booth**

Conclusion

- Presented a set of tools for top-down planning of interconnect using bus topologies
- Assisted bus planning provides:
 - A rich set of good topologies to choose from
 - Runtime: ~5000 topologies per minute
 - Hierarchical design w/o need for terminals
- Auto bus planning provides:
 - Fast techniques to legalize topologies
 - Runtime: 25k nets per minute
 - Fast and accurate analysis of congestion
 - Enables accurate timing analysis early on

Q&A